

Statistical and Computational Models

Lecture 9: Ordered and unordered outcomes

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Roadmap

Ordered outcomes

Unordered outcomes

Irrelevant alternatives and extensions

More than two categories

Very often we encounter outcomes with more than two categories:

$$Y_i \in \{1, 2, \dots, J\}$$

Examples in political science:

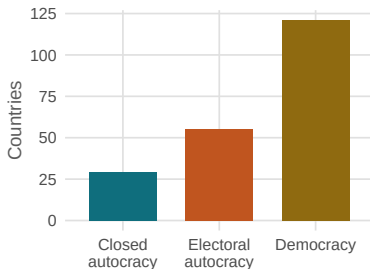
- Survey items (strongly disagree, ..., strongly agree)
- Regime type (closed autocracy, electoral autocracy, democracy)
- Vote choice (no vote, party A, ..., party J)

Note the difference between **ordinal** (e.g. regime type) and **nominal** (e.g. vote choice) variables: the first has an **order but no known spacing**, the second **no order at all**.

More than two categories

Suppose our running example actually had three regime types:

regime	y_num	lgdp	oil
Democracy	3	9.23	0
Democracy	3	8.32	0
Closed autocracy	1	6.49	1
Electoral autocracy	2	7.56	1



We want to model regime type – now **ordinal** – with the same covariates as before, `oil` and `lgdp`. Why not just use the models we already know?

The models we already have

LEAST SQUARES

Treat `regime` as a continuous variable (`y_num`):

$$\widehat{y_num}_i = -1.11 \\ + 0.41 \lgdp_i - 0.33 \text{ oil}_i$$

Assumes the regimes are **equally spaced**. Fitted values run from 0.82 to 3.63.

We want a model that uses the order (like OLS), but assumes nothing about spacing (like logit).

LOGIT

Compare democracy against all alternatives:

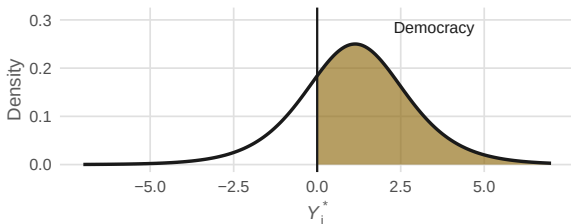
$$\text{logit } \hat{\mathbb{P}}(\text{dem.}_i) = -15.33 \\ + 1.85 \lgdp_i - 1.56 \text{ oil}_i$$

Ignores the qualitative **ordering** of closed and electoral autocracies.

Latent variable interpretation

Suppose an unobserved **latent** propensity to be a democracy, Y_i^* :

$$Y_i^* = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \sim \text{Logistic}(0, 1), \quad Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

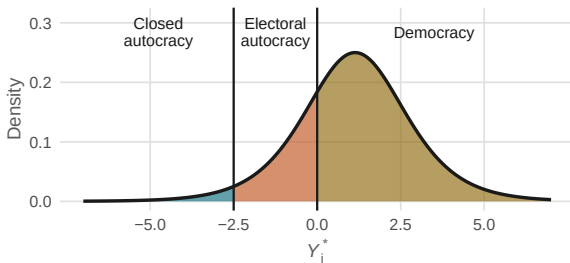


For a given country, Y_i^* follows a **logistic** density centred on $\mathbf{x}_i^\top \boldsymbol{\beta}$. We only observe Y_i , which records whether Y_i^* fell above the cutpoint.

Apply logit^{-1} to $\mathbf{x}_i^\top \boldsymbol{\beta}$ and we get $\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)$ – the **shaded area**.

Latent variable with multiple cutpoints

Now add a **second cutpoint**, below the first:



Each category's probability is now the **area between two cutpoints**.

Ordered logit

$Y_i^* = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ gives rise to Y_i through the cutpoints:

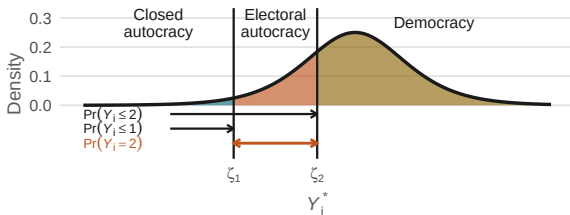
$$Y_i = \begin{cases} 1 & \text{if } Y_i^* \leq \zeta_1 \\ 2 & \text{if } \zeta_1 < Y_i^* \leq \zeta_2 \\ 3 & \text{if } Y_i^* > \zeta_2 \end{cases}$$

where ζ_1, ζ_2 are **cutpoints** (a.k.a. thresholds).

So we have **two things to estimate**: the $\boldsymbol{\beta}$, which linearly map $\mathbf{x}_i$ to the latent scale Y_i^* , and the ζ s, which cut that scale into the categories of Y_i (here, regimes).

Can we estimate an intercept *and* both ζ s? Why not?

Ordered logit



Generally, the area left of cutpoint j is the **ordered logit**:

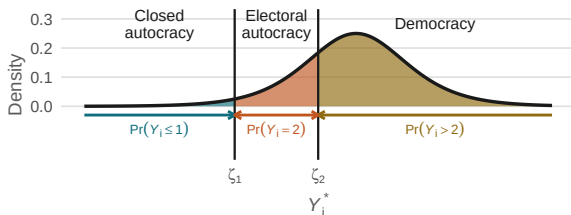
$$\mathbb{P}(Y_i \leq j \mid \mathbf{x}_i) = \text{logit}^{-1}(\zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta})$$

Category j 's probability is the area left of the j th cutpoint, minus the area left of the previous, $(j - 1)$ th, cutpoint:

$$\mathbb{P}(Y_i = j \mid \mathbf{x}_i) = \text{logit}^{-1}(\zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta}) - \text{logit}^{-1}(\zeta_{j-1} - \mathbf{x}_i^\top \boldsymbol{\beta})$$

Together, $\boldsymbol{\beta}$ and the ζ s tie $\mathbf{x}_i$ to the probability of each category.

Ordered logit



For our three regimes, each probability is its area under the density:

$$\mathbb{P}(\text{closed}) = \text{logit}^{-1}(\zeta_1 - \mathbf{x}_i^\top \boldsymbol{\beta})$$

$$\mathbb{P}(\text{electoral}) = \text{logit}^{-1}(\zeta_2 - \mathbf{x}_i^\top \boldsymbol{\beta}) - \text{logit}^{-1}(\zeta_1 - \mathbf{x}_i^\top \boldsymbol{\beta})$$

$$\mathbb{P}(\text{democracy}) = 1 - \text{logit}^{-1}(\zeta_2 - \mathbf{x}_i^\top \boldsymbol{\beta})$$

Fitting ordered logit in R

Using `MASS::polr()`:

```
fit_ologit <- MASS::polr(regime ~ lgdp + oil, data = countries,  
                        Hess = TRUE)  
round(coef(summary(fit_ologit))[, 1:2], 2)
```

	Value	Std. Error
lgdp	1.96	0.24
oil	-1.54	0.37
Closed autocracy Electoral autocracy	13.75	1.89
Electoral autocracy Democracy	16.30	2.03

The first two rows, `lgdp` and `oil`, are the **coefficients** $\hat{\beta}$, which linearly map $\mathbf{x}_i$ to the latent scale Y_i^* .

The last two rows are the **cutpoints** $\hat{\zeta}_1, \hat{\zeta}_2$, one per boundary between adjacent regimes. There is no intercept row: it is absorbed into the $\hat{\zeta}$ s.

What is β in ordered logit?

Consider a single β_m inside our vector β :

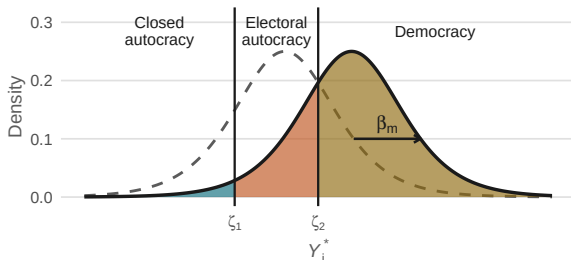
$$\beta_m = \underbrace{\frac{\partial}{\partial x_{im}}}_{\text{change in } x_{im}} \log \underbrace{\frac{\mathbb{P}(Y_i > j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i \leq j \mid \mathbf{x}_i)}}_{\text{odds of being above cutpoint } j}$$

Read: how far the log odds of being **above** cutpoint j move per unit of x_{im} – the same for every j (**proportional odds**).

1. Read the **sign**: **oil** moves countries down the order, **lgdp** up – but only the **first and last** categories must move that way. Why?
2. Exponentiate: switching **oil** on multiplies the odds of being above **any** cutpoint by $e^{\hat{\beta}_{\text{oil}}} = 0.21$.
3. For a change of size c , $e^{c\hat{\beta}_m}$. Doubling GDP per capita: **3.9**.

Note: all three are about **cumulative** odds, not any one category.

Shifting the density



Increase x_{im} by one unit: the whole density slides right by β_m (dashed: before). The cutpoints do not move, so the log odds of being above **every** cutpoint rise by β_m .

So how does the middle area – the probability of electoral autocracy – change?

Predicted probabilities

Recall that the **predicted probabilities** of each regime, for each country, can be recovered by multiplying the model matrix by $\hat{\beta}$ to get $\mathbf{x}_i^\top \hat{\beta}$, then plugging it and the $\hat{\zeta}$ s into the ordered logit:

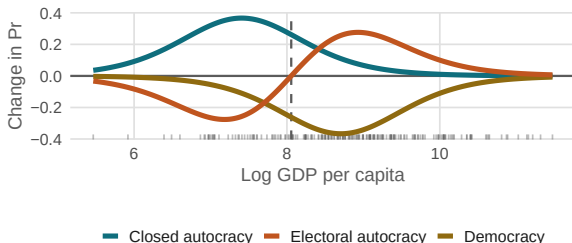
Observed	lgdp	oil	Closed	Electoral	Democracy
Democracy	9.23	0	0.01	0.13	0.86
Democracy	8.32	0	0.07	0.42	0.51
Closed autocracy	6.49	1	0.93	0.07	0.01
Electoral autocracy	7.56	1	0.62	0.34	0.05

Each row sums to one. In R, `predictions()` from `{marginaleffects}` returns them, one row per country per regime.

How does each column move when `oil` switches on?

Interpreting oil, *ceteris paribus*

Switch oil on: the change in fitted Pr, regime by regime:



Dashed: where oil's effect on electoral autocracy changes sign.

Note that $\mathbf{x}_i^\top \boldsymbol{\beta}$ appears on both sides of the minus sign in $\mathbb{P}(Y_i = j \mid \mathbf{x}_i)$: the direction is **ambiguous**, except at the top and bottom.

Interpreting oil, *ceteris paribus*

Averaging each curve over the observed countries gives one **APE** per regime:

```
margineffects::avg_comparisons(fit_ologit, variables = "oil")
```

	Group	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
Closed autocracy		0.1269	0.0332	3.83	<0.001	12.9	0.0619	0.192
Electoral autocracy		0.0911	0.0216	4.21	<0.001	15.3	0.0487	0.133
Democracy		-0.2180	0.0501	-4.35	<0.001	16.1	-0.3162	-0.120

Term: oil

Type: probs

Comparison: 1 - 0

The three APEs sum to zero. Why?

Each country's probabilities sum to one, with or without oil: whatever oil adds to one category must come out of the others.

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Outcomes without an order

What if our autocracies came in three **categories**: military, single-party and personalist?

There is **no order** here, only qualitative differences. Can we model this outcome as a function of $\mathbf{x}_i$?

For each unit, each category has its own latent propensity, Y_{ij}^* , governed by a category-specific β_j . Whichever is **largest** determines Y_i .

With independent Gumbel (extreme-value) errors, this gives the **multinomial logit**.

The multinomial logit

$$\mathbb{P}(Y_i = j \mid \mathbf{x}_i) = \frac{\overbrace{\exp(\mathbf{x}_i^\top \beta_j)}^{\text{category } j\text{'s score}}}{\underbrace{\sum_{k=1}^J \exp(\mathbf{x}_i^\top \beta_k)}_{\text{all } J \text{ scores, } j \text{ included}}}, \quad \beta_1 = \mathbf{0}$$

Read: exponentiating $\mathbf{x}_i^\top \beta_j$ gives each category a positive **score**. The probability of category j is its **share** of all J scores.

Each category j has **its own** vector $\beta_j = (\beta_{j0}, \beta_{j1}, \dots, \beta_{jp})^\top$: an intercept and one slope per covariate. Note that $\sum_j \mathbb{P}(Y_i = j \mid \mathbf{x}_i) = 1$. Why?

Only **differences** between scores matter: shift every β_j by the same amount and the shares do not change. So we pin one, the **base category**, at $\beta_1 = \mathbf{0}$: its score is $e^0 = 1$.

With $J = 2$ this is the logit. The cost: $(J - 1)(p + 1)$ parameters.

What is β in the multinomial logit?

Consider a single β_{jm} , the slope on x_{im} in category j 's equation:

$$\beta_{jm} = \underbrace{\frac{\partial}{\partial x_{im}}}_{\text{change in } x_{im}} \log \underbrace{\frac{\mathbb{P}(Y_i = j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)}}_{\text{odds of } j \text{ relative to the base, category 1}}$$

Read: how the log odds of j **relative to the base** category move.

1. Read the **sign**: relative to the base only.
2. Exponentiate: $e^{\hat{\beta}_{jm}}$ multiplies $\mathbb{P}(Y_i = j \mid \mathbf{x}_i) / \mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)$.
3. For a change of size c , $e^{c\hat{\beta}_{jm}}$, as before.

Note: none of the three speaks to $\mathbb{P}(Y_i = j \mid \mathbf{x}_i)$ itself.

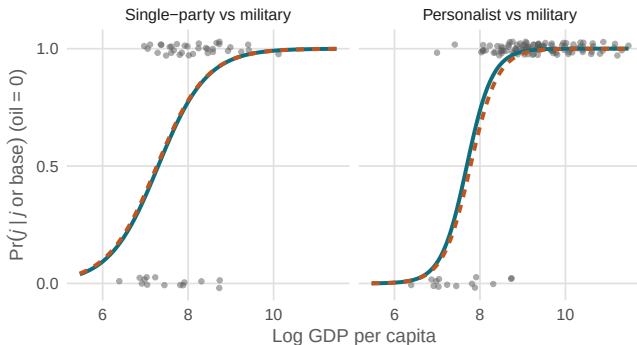
A series of logits

Divide any category's probability by the base's, and the denominator cancels:

$$\log \frac{\mathbb{P}(Y_i = j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i)} = \mathbf{x}_i^\top \beta_j$$

So each β_j is a **binary logit**: among countries in **category j or the base**, the chance of being in j .

A series of logits



Solid: the multinomial logit. Dashed: a separate `glm()` on each pair.

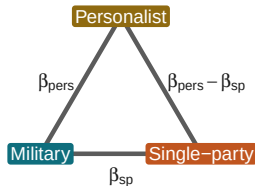
These are both logits, but estimated differently. The multinomial fits them **jointly**: more precise, and coherent across every pair of categories – if the model is right.

The comparisons must add up

Three categories allow three pairwise logits – but personalist vs single-party is just $\beta_{\text{pers}} - \beta_{\text{sp}}$, the difference of the other two.

oil's coefficient in each comparison:

Comparison	Separate	Multinomial
Single-party vs military	-1.02	-0.94
Personalist vs military	-1.59	-2.37
Personalist vs single-party	-1.51	-1.43
Second row minus first	-0.56	-1.43



The multinomial's comparisons **add up**; three separate logits need not.

Similar when the model is right and each pair has plenty of data.

Dissimilar when pairs are small, or the model is wrong: e.g. the categories are ordered, or some are close substitutes.

What oil does

Fitting it in R using `nnet::multinom()`, military as the base:

```
fit_mnlogit <- nnet::multinom(autocracy ~ lgdp + oil, data = countries,  
                             trace = FALSE)  
round(coef(fit_mnlogit), 2)
```

	(Intercept)	lgdp	oil
Single-party	-12.78	1.75	-0.94
Personalist	-25.68	3.34	-2.37

And here are the APEs from `avg_comparisons(fit_mnlogit, variables = "oil", type = "probs")`:

Category	Coefficient on oil	APE of oil on Pr
Military	–	0.101
Single-party	-0.938	0.128
Personalist	-2.372	-0.229

A negative coefficient, a **positive** effect on the probability! Why?

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Irrelevant alternatives

The multinomial logit assumes the unobserved factors behind one category's latent propensity are **unrelated** to those behind another's. When is that plausible?

Commuters split evenly between a car and a red bus. Now add a blue bus, identical but for its colour (shares in %):

Option	Before	After, under IIA	After, more likely
Car	50	33	50
Red bus	50	33	25
Blue bus	–	33	25

The model has the blue bus take riders from the car and the red bus in **proportion**. Plausible? Probably not...

Irrelevant alternatives

This is **independence of irrelevant alternatives** (IIA):

$$\frac{\mathbb{P}(Y_i = j \mid \mathbf{x}_i)}{\mathbb{P}(Y_i = k \mid \mathbf{x}_i)} = \exp(\mathbf{x}_i^\top (\beta_j - \beta_k))$$

Read: the ratio of two categories' probabilities ignores every other category, however close a substitute.

Relaxing IIA

Three ways to let the blue bus take its riders from the **red bus**, not the car:

Model	Idea	The buses
Nested logit	Group similar options into nests; IIA holds only within a nest	{car}, {red bus, blue bus}: a new bus draws from its own nest
Multinomial probit	Let the unobserved parts of the utilities be correlated	Whatever makes someone like one bus makes them like the other
Mixed logit	Let tastes vary across people (random coefficients)	Bus lovers and car lovers: a new bus draws from the bus lovers

All three are in `{mlogit}` (won't cover). They work best with covariates that vary by option, like fares or travel times. With only country-level covariates like ours, their extra parameters are weakly identified.

References

- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – **ch. 6**. grodr.github.io/glms.
- Bürkner, P.-C. & Vuorre, M. (2019) "Ordinal Regression Models in Psychology: A Tutorial." *Advances in Methods and Practices in Psychological Science* 2(1): 77–101.